POSTER REVIEW

Critique of "The Five-Step Process of Hydrothermal Buffering" Poster

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1. Figure Reviewed

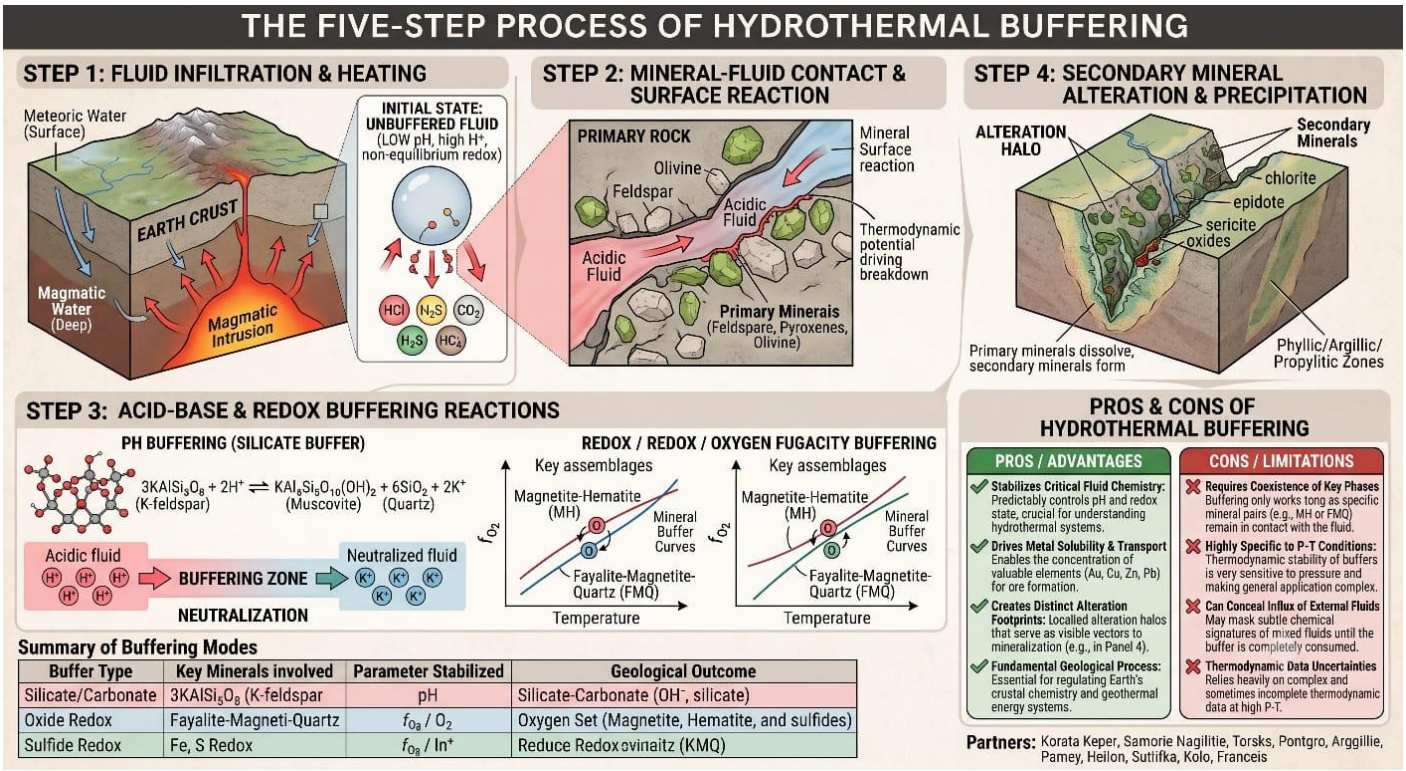


Figure 1: Original poster submitted for review.

2. Overview

The poster explains hydrothermal buffering — the process by which acidic, magmatic-derived fluids are neutralized and re-equilibrated in oxygen fugacity as they react with primary crustal minerals — across a set of numbered panels, a summary table, and a Pros/Cons box. The chemistry and overall concept are sound, but the poster was compressed from a longer, ten-step version of the process into five steps, and that compression introduced the sequencing and content gaps documented below.

3. Identified Flaws & Recommended Solutions

#	Flaw Observed	Why It Is a Problem	Recommended Fix
1	Poster is titled a "Five-Step Process" but only Steps 1, 2, 3, and 4 are labelled anywhere on the panel — no Step 5 exists.	Breaks the promise made in the headline; a reader actively looks for a fifth box and assumes something is missing or mislabelled.	Add an explicit Step 5 panel: "Buffer Exhaustion & Fluid Breakthrough" — the point where the mineral buffer is consumed and the fluid reverts to an unbuffered, reactive state. This also gives the Cons box ("buffer only works as long as mineral pairs

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			remain") a natural home on the diagram instead of being an orphaned bullet.
2	Reading order does not follow the numbers: Step 4 sits in the top-right, ahead of Step 3, which is placed below Steps 1–2.	Viewers scan posters left-to-right, top-to-bottom by default; jumping 1→2→4→3 forces re-reading and hides the causal chain.	Re-lay the grid so panels run in strict numeric order (row 1: Steps 1–2–3, row 2: Steps 4–5), and add small connector arrows/numbers between panels to make the sequence unambiguous.
3	The redox panel heading reads "REDOX / REDOX / OXYGEN FUGACITY BUFFERING" and shows two nearly identical fO_2-vs-temperature graphs.	"Redox" is repeated, and the duplicate graph adds no new information while eating poster space that could go to the missing Step 5.	Merge into one heading ("Redox / Oxygen Fugacity Buffering") and one graph with both the Magnetite-Hematite and Fayalite-Magnetite-Quartz buffer curves plotted together.
4	Muscovite is written as $KAl_6Si_5O_{10}(OH)_2$ in the pH-buffering reaction.	The correct formula is $KAl_3Si_3O_{10}(OH)_2$. On a technical/scientific poster, an incorrect mineral formula undermines credibility and can fail a factual review.	Correct the formula and re-check that the full reaction balances for charge and mass before finalizing.
5	Several typos in the summary table: "fO_8" (should be fO_2), "In^+" (should be H^+), "Fayalite-Magneti-Quartz" (missing 'te'), and "Reduce Redox ovinaitz (KMQ)", which is not legible text.	Typos in a summary/reference table are highly visible and suggest the poster was not proofread — this is often the first thing a grader marks down.	Proofread the table against the source notes; rewrite the garbled cell as something like "Sets fO_2 via the FMQ/KMQ buffer."
6	The alteration-halo cross-section (Step 4) labels secondary minerals and zones (chlorite, epidote, sericite, oxides, Phyllic/Argillic/Propylitic) but has no legend tying colors to zone names.	Without a legend, the color-coding is decorative rather than informative, so readers cannot actually map a color on the block diagram to a named zone.	Add a compact color-key legend beside the diagram, matching each zone color to its label.
7	The Pros/Cons panel uses dense, multi-line paragraph bullets in a small font.	Poster text is normally read from 1–2 m away; long bulleted paragraphs at small point size are the hardest element on the page to read at that distance.	Shorten each bullet to a single line (bold keyword + one short clause), and let the sentence detail move to a caption or an appendix/notes section instead.
9	No source citations for the thermodynamic buffer curves (MH, FMQ) or mineral data.	Scientific posters are expected to credit the thermodynamic datasets/diagrams they are built from; omitting this weakens the poster's academic rigor.	Add a small "Sources" line (e.g., dataset/reference used for the fO_2-T buffer curves) in a corner of the panel.

4. Note on Reducing the Process from 10 Steps to 5

Condensing ten steps into five only works if every original step is deliberately folded into one of the five panels; right now four panels are visible and no panel accounts for the end of the buffering process (when the mineral buffer is used up). The cleanest fix is to treat the five panels as:

- Step 1 – Fluid infiltration & heating (unbuffered fluid formed)

- Step 2 – Mineral–fluid contact & surface reaction
- Step 3 – Acid-base and redox buffering reactions
- Step 4 – Secondary mineral alteration & precipitation
- Step 5 – Buffer exhaustion & fluid breakthrough (new panel)

This keeps the five-panel promise in the title, restores a clean 1→2→3→4→5 reading order, and gives the poster's own "Cons" point about buffer capacity a proper place in the process diagram rather than leaving it stranded in a side box.

5. Summary

Most of the issues above are quick fixes — a missing panel, a mislabeled formula, a duplicated graph, and a handful of typos — rather than problems with the underlying science, which is accurate. Addressing the nine items in Section 3 should bring the poster fully in line with its own five-step title and make it easier to read at poster distance.

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